27/01/2024
Evening

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2024 (Online) Phase-1

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 90 questions. Each subject (MPC) has 30 questions. The maximum marks are 300.

(3) This question paper contains **Three Parts**. **Part-A** is Mathematics, **Part-B** is Physics and **Part-C** is **Chemistry**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt any 05 questions out of 10 Questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 30) contains** 10 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. Let e_1 be the eccentricity of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ and e_2 be the eccentricity of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a > b$, which passes through the foci of the hyperbola. If $e_1e_2 = 1$, then the length of the chord of the ellipse parallel to the x-axis and passing through (0, 2) is:

(1) $\frac{10\sqrt{5}}{3}$ (2) $3\sqrt{5}$

(3) $4\sqrt{5}$ (4) $\frac{8\sqrt{5}}{3}$

Answer (1)

Sol. $\frac{x^2}{16}-\frac{y^2}{9}=1$

$e_1=\frac{\sqrt{16+9}}{4}=\frac{5}{4}$

foci (± 5, 0)

$e_1e_2 = 1$

$\therefore \quad e_2=\frac{4}{5}$

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ passes through (± 5, 0)

$\therefore \quad \frac{25}{a^2}=1\Rightarrow a^2=25$

$e_2=\frac{\sqrt{a^2-b^2}}{a}$

$\frac{4}{5}=\frac{\sqrt{25-b^2}}{5}$

$\Rightarrow \quad b = 3$

$\frac{x^2}{25}+\frac{y^2}{9}=1$

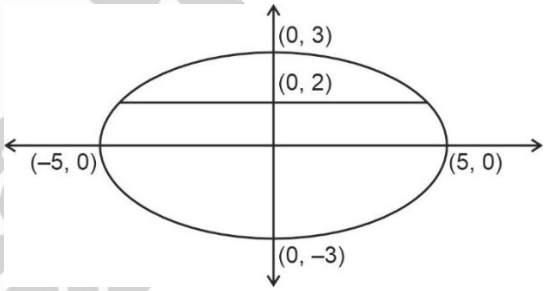
For $y = 2$

$\frac{x^2}{25}+\frac{4}{9}=1$

$\Rightarrow \quad \frac{x^2}{25}=1-\frac{4}{9}=\frac{5}{9}$

$\Rightarrow \quad x^2=\frac{25\times5}{9}\Rightarrow x=\frac{5\sqrt{5}}{3}$

$\therefore$ Length of chord = $2\times\frac{5\sqrt{5}}{3}=\frac{10\sqrt{5}}{3}$



2. The values of α, for which

$$\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix}=0$$, lie in the interval

(1) (–2, 1) (2) (0, 3)

(3) (–3, 0) (4) $\left(-\frac{3}{2},\frac{3}{2}\right)$

Answer (3)



Sol. $\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix} = 0$

$C_3 \to C_3 - (\alpha C_1 + C_2)$

$\Rightarrow \begin{vmatrix} 1 & \frac{3}{2} & 0 \\ 1 & \frac{1}{3} & 0 \\ 2\alpha+3 & 3\alpha+1 & -(2\alpha^2+6\alpha+1) \end{vmatrix} = 0$

$\Rightarrow 2\alpha^2 + 6\alpha + 1 = 0$

$\alpha = \frac{-3+\sqrt{7}}{2}, \frac{-3-\sqrt{7}}{2}$

3. Let R be the interior region between the lines $3x - y + 1$ and $x + 2y - 5 = 0$ containing the origin. The set of all values of a, for which the points $(a^2, a + 1)$ lie in R, is:

(1) $(-3, 0) \cup \left(\frac{2}{3}, 1\right)$ (2) $(-3, -1) \cup \left(\frac{1}{3}, 1\right)$

(3) $(-3, -1) \cup \left(\frac{1}{3}, 1\right)$ (4) $(-3, 0) \cup \left(\frac{1}{3}, 1\right)$

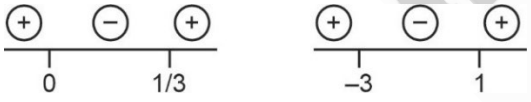
Answer (4)

Sol. $(a^2, a + 1)$ and $(0, 0)$ lies on the same side of $3x - y + 1 = 0$ and $x + 2y - 5 = 0$

$\Rightarrow 3a^2 - (a + 1) + 1 > 0$ and $a^2 + 2a + 2 - 5 < 0$

$\Rightarrow a(3a - 1) > 0$ and $a^2 + 2a - 3 < 0$

$\Rightarrow a(3a - 1) > 0$ and $(a + 3)(a - 1) < 0$



$\Rightarrow a \in (-\infty, 0) \cup \left(\frac{1}{3}, \infty\right)$ and $a \in (-3, 1)$

$\Rightarrow a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$

So, option (4) is correct.

4. The integral $\int \frac{(x^8 - x^2)dx}{(x^{12} + 3x^6 + 1)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$ is equal to:

(1) $\log_e \left(\left| \tan^{-1}\left(x^3 + \frac{1}{x^3}\right) \right| \right) + C$

(2) $\log_e \left(\left| \tan^{-1}\left(x^3 + \frac{1}{x^3}\right) \right| \right)^3 + C$

(3) $\log_e \left(\left| \tan^{-1}\left(x^3 + \frac{1}{x^3}\right) \right| \right)^{1/3} + C$

(4) $\log_e \left(\left| \tan^{-1}\left(x^3 + \frac{1}{x^3}\right) \right| \right)^{1/2} + C$

Answer (3)

Sol. Let $\tan^{-1}\left(x^3 + \frac{1}{x^3}\right) = t$

$\Rightarrow \frac{3}{1 + \left(x^3 + \frac{1}{x^3}\right)^2}\left(x^2 - \frac{1}{x^4}\right)dx = dt$

$\Rightarrow \frac{3(x^8 - x^2)}{x^{12} + 3x^6 + 1}dx = dt$

$\int \frac{1}{3}\frac{dt}{t} = \frac{1}{3}\ln|t| + C$

$= \frac{1}{3}\ln\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right| + C$

$= \log_e \left(\left| \tan^{-1}\left(x^3 + \frac{1}{x^3}\right) \right| \right)^{1/3} + C$

Hence, option (3) is correct.

5. If α, β are the roots of the equation $x^2 - x - 1 = 0$ and $S_n = 2023\,\alpha^n + 2024\,\beta^n$, then:

(1) $S_{11} = S_{10} + S_{12}$ (2) $2S_{11} = S_{12} + S_{10}$

(3) $2S_{12} = S_{11} + S_{10}$ (4) $S_{12} = S_{11} + S_{10}$

Answer (4)



Sol. We have, $x^2 - x - 1 = 0$

By Newton's method

$S_n - S_{n-1} - S_{n-2} = 0$

$\therefore \quad S_n = S_{n-1} + S_{n-2}$

$\therefore \quad S_{12} = S_{11} + S_{10}$

6. Consider the function $f : (0, 2) \to \mathbb{R}$ defined by $f(x) = \frac{x}{2} + \frac{2}{x}$ and the function $g(x)$ defined by $g(x) = \begin{cases} \min\{f(t)\} & , \ 0 < t \le x \text{ and } 0 < x \le 1 \\ \frac{3}{2} + x & , \ 1 < x < 2 \end{cases}$. Then,

(1) g is not continuous for all $x \in (0, 2)$

(2) g is continuous and differentiable for all $x \in (0, 2)$

(3) g is continuous but not differentiable at $x = 1$

(4) g is neither continuous nor differentiable at $x = 1$

Answer (3)

Sol. $f(x) = \frac{x}{2} + \frac{2}{x}$

$\Rightarrow \quad f'(x) = \frac{1}{2} - \frac{2}{x^2} = \left(\frac{x^2 - 4}{2x^2}\right)$ for $x \in (0, 2)$

$\Rightarrow \quad f'(x) < 0 \Rightarrow f(x)$ is decreasing

$g(x) = \begin{cases} \min\{f(t)\} & , \ 0 < t \le x \text{ and } 0 < x \le 1 \\ \frac{3}{2} + x & , \ 1 < x < 2 \end{cases}$

Since $f(x)$ is decreasing, $f(t)$ is minimum for x in $t \in (0, x] \Rightarrow \min\{f(t)\} = f(x)$

$\Rightarrow \quad g(x) = \begin{cases} f(x) & , \ x \in (0, 1] \\ \frac{3}{2} + x & , \ 1 < x < 2 \end{cases}$

$\Rightarrow \quad g(1) = \frac{1}{2} + \frac{2}{1} = \frac{5}{2}$ also $g(1^+) = \frac{3}{2} + 1 = \frac{5}{2}$

$\Rightarrow \quad g(x)$ is continuous in $(0, 2)$

$g'(x) = \begin{cases} f'(x) & , \ x \in (0, 1] \\ 1 & , \ x \in (1, 2) \end{cases}$

$g'(1) = f'(1) = \frac{1-4}{2} = \frac{-3}{2}$

$\Rightarrow \quad g'(1) \ne g'(1^+) \Rightarrow$ not differentiable at $x = 1$

7. Let $f : \mathbb{R} - \left\{\frac{-1}{2}\right\} \to \mathbb{R}$ and $g : \mathbb{R} - \left\{\frac{-5}{2}\right\} \to \mathbb{R}$ be defined as $f(x) = \frac{2x+3}{2x+1}$ and $g(x) = \frac{|x|+1}{2x+5}$. Then, the domain of the function fog is:

(1) $\mathbb{R}$ (2) $\mathbb{R} - \left\{-\frac{5}{2}\right\}$

(3) $\mathbb{R} - \left\{-\frac{7}{4}\right\}$ (4) $\mathbb{R} - \left\{-\frac{5}{2}, -\frac{7}{4}\right\}$

Answer (2)

Sol. $f(g(x))$

$\Rightarrow g(x) \ne -\frac{1}{2}$

$\frac{11+1}{2x+5} \ne -\frac{1}{2}$

Case I : $x \ge 0$

$\frac{x+1}{2x+5} = \frac{-1}{2}$

$2x + 2 = -2x - 5$

$4x = -7$

$x = -\frac{7}{4}$ (Rejected)

Case II: $x < 0$

$\frac{-x+1}{2x+5} = \frac{-1}{2}$

$-2x + 2 = -2x - 5$

$2 = -5$ (Not possible)

$\Rightarrow$ Domain of $f(g(x))$ = domain of $g(x)$

$D_{fog} = \mathbb{R} - \left\{-\frac{5}{2}\right\}$



8. For $0 < a < 1$, the value of the integral $\int_0^{\pi} \frac{dx}{1-2a\cos x+a^2}$ is:

(1) $\frac{\pi}{\pi+a^2}$ (2) $\frac{\pi}{1+a^2}$

(3) $\frac{\pi}{1-a^2}$ (4) $\frac{\pi^2}{\pi-a^2}$

Answer (3)

Sol. Let $I = \int_0^{\pi} \frac{dx}{1+a^2-2a\cos x}$

$$I = \frac{1}{2a}\int_0^{\pi} \frac{dx}{\frac{1+a^2}{2a}-\cos x}$$

Let $\alpha = \frac{1+a^2}{2a}$

$$I = \frac{1}{2a}\int_0^{\pi} \frac{dx}{\alpha-\cos x}$$

$$= \frac{1}{2a}\int_0^{\pi} \frac{dx}{\alpha-\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}}$$

$$= \frac{1}{2a}\int_0^{\pi} \frac{\sec^2\frac{x}{2}dx}{(\alpha-1)+(\alpha+1)\tan^2\frac{x}{2}}$$

Let $\tan\frac{x}{2} = t \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx = dt$

$$I = \frac{1}{a}\int_0^{\infty} \frac{dt}{(\alpha-1)+(\alpha+1)t^2}$$

$$= \frac{1}{a(\alpha+1)}\int_0^{\infty} \frac{dt}{\left(\frac{\alpha-1}{\alpha+1}\right)+t^2}$$

$$= \frac{1}{a(\alpha+1)}\frac{1}{\sqrt{\frac{\alpha-1}{\alpha+1}}}\tan^{-1}\left(\frac{t}{\sqrt{\frac{\alpha-1}{\alpha+1}}}\right)\Bigg|_0^{\infty}$$

$$= \frac{1}{a\sqrt{\alpha^2-1}}\left[\tan^{-1}\infty-\tan^{-1}0\right]$$

$$= \frac{\pi}{2a\sqrt{\alpha^2-1}} = \frac{\pi}{2a\sqrt{\left(\frac{1+a^2}{2a}\right)^2-1}} = \frac{\pi}{\sqrt{\left(1-a^2\right)^2}}$$

$$= \frac{\pi}{\left(1-a^2\right)} = \frac{\pi}{1-a^2}$$

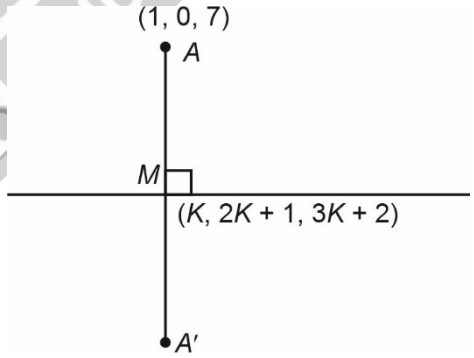
9. Let the image of the point (1, 0, 7) in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ be the point (α, β, γ). Then which one of the following points lies on the line passing through (α, β, γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with *y*-axis and *z*-axis respectively and an acute angle with *x*-axis?

(1) $(3, 4, 3-2\sqrt{2})$ (2) $(1, 2, 1-\sqrt{2})$

(3) $(3, -4, 3+2\sqrt{2})$ (4) $(1, -2, 1+\sqrt{2})$

Answer (1)

Sol.



$$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} = K$$



$\overrightarrow{AM} = (K-1)\hat{i} + (2K+1)\hat{j} + (3K-5)\hat{k}$

$\overrightarrow{AM} \perp (\hat{i} + 2\hat{j} + 3\hat{k})$

$\Rightarrow (K-1) + 2(2K+1) + 3(3K-5) = 0$

$K + 4K + 9K - 1 + 2 - 15 = 0$

$14K = 14 \Rightarrow K = 1$

$A' \equiv (1, 6, 3)$

Line passing through (1, 6, 3) and angle

$\left(\frac{2\pi}{3}, \frac{3\pi}{4}, \theta\right), \theta \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow \cos^2\left(\frac{2\pi}{3}\right) + \cos^2\left(\frac{3\pi}{4}\right) + \cos^2\theta = 1$

$\Rightarrow \frac{1}{4} + \frac{1}{2} + \cos^2\theta = 1$

$\Rightarrow \cos^2\theta = \frac{1}{4} \Rightarrow \cos\theta = \frac{1}{2}$

$\Rightarrow \frac{x-1}{\frac{1}{2}} = \frac{y-6}{-\frac{1}{2}} = \frac{z-3}{-\frac{1}{\sqrt{2}}}$

10. The 20th term from the end of the progression $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, ..., -129\frac{1}{4}$ is :

(1) –118 (2) –100

(3) –115 (4) –110

Answer (3)

Sol. Given $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, ..., -129\frac{1}{4}$

$-\frac{517}{4} = 20 + (n-1)\left(-\frac{3}{4}\right)$

$\Rightarrow$ –517 = 80 + (–3n + 3)

$\Rightarrow$ –597 = –3n + 3

$\Rightarrow$ –600 = –3n

$\Rightarrow$ n = 200

n^{th} term from end is $(n - r + 1)^{th}$

$\therefore$ 200 – 20 + 1 = 181st term

$a_{181} = 20 + (181 - 1)\left(-\frac{3}{4}\right)$

$a_{181} = 20 - 135$

$a_{181} = -115$

11. Let $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ and $f''(x) > 0$ for all $x \in (0, 3)$. If g is decreasing in $(0, \alpha)$ and increasing in $(\alpha, 3)$, then 8α is:

(1) 24 (2) 18

(3) 20 (4) 0

Answer (2)

Sol. $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$

$g'(x) = f'\left(\frac{x}{3}\right) + f'(3-x)(-1)$

Putting $g'(x) = 0$

$f'\left(\frac{x}{3}\right) = f'(3-x)$

$\Rightarrow \frac{x}{3} = 3 - x$

$\Rightarrow x = 9 - 3x$

$\Rightarrow 4x = 9$

$\Rightarrow x = \frac{9}{4}$

As g is decreasing in $(0, \alpha)$

$f'\left(\frac{x}{3}\right) < f'(3-x)$ in $(0, \alpha)$

g is increasing in $(\alpha, 3)$

$f'\left(\frac{x}{3}\right) > f'(3-x)$ in $(\alpha, 3)$

$\Rightarrow \alpha = \frac{9}{4}$

$8\alpha = 8 \times \frac{9}{4}$

= 18



12. The position vectors of the vertices A, B and C of a triangle are $2\hat{i}-3\hat{j}+3\hat{k}, 2\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+\hat{j}+3\hat{k}$ respectively. Let l denotes the length of the angle bisector AD of $\angle BAC$ where D is on the line segment BC, then $2l^2$ equals:

(1) 42 (2) 50

(3) 45 (4) 49

Answer (3)

Sol. Position vectors are

$A(2, -3, 3)$, $B(2, 2, 3)$, $C(-1,1,3)$

$\overrightarrow{AB} = \vec{B} - \vec{A} = (2\hat{i}+2\hat{j}+3\hat{k}) - (2\hat{i}-3\hat{j}+3\hat{k})$

$= 5\hat{j}$

$\overrightarrow{AC} = \vec{C} - \vec{A} = (-\hat{i}+\hat{j}+3\hat{k}) - (2\hat{i}-3\hat{j}+3\hat{k})$

$= 3\hat{i}+4\hat{j}$

$|\overrightarrow{AB}| = 5$

$|\overrightarrow{AC}| = 5$

Bisector of angle A meets $\overrightarrow{BC}$ at D

So $\overrightarrow{AD}$ divides $\overrightarrow{BC}$ in the ratio $\overrightarrow{AB} : \overrightarrow{AC}$

Position vector of D

$$= \frac{|\overrightarrow{AC}|(2\hat{i}+2\hat{j}+3\hat{k}) + |\overrightarrow{AB}|(-\hat{i}+\hat{j}+3\hat{k})}{|\overrightarrow{AB}| + |\overrightarrow{AC}|}$$

$$= \frac{5(2\hat{i}+2\hat{j}+3\hat{k}) + 5(-\hat{i}+\hat{j}+3\hat{k})}{5+5}$$

$$= \frac{10\hat{i}+10\hat{j}+15\hat{k}-5\hat{i}+5\hat{j}+15\hat{k}}{10}$$

$$= \frac{5\hat{i}+15\hat{j}+30\hat{k}}{10}$$

So,

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \frac{5\hat{i}+15\hat{j}+30\hat{k}}{10} - (2\hat{i}-3\hat{j}+3\hat{k})$$

$$= \frac{5\hat{i}+15\hat{j}+30\hat{k}-20\hat{i}+30\hat{j}-30\hat{k}}{10}$$

$$= \frac{-15\hat{i}+45\hat{j}}{10}$$

$$2l^2 = \frac{4500}{100}$$

$2l^2 = 45$

13. If $\lim\limits_{x\to 0} \dfrac{3+\alpha\sin x+\beta\cos x+\log_e(1-x)}{3\tan^2 x} = \dfrac{1}{3}$, then $2\alpha - \beta$ is equal to:

(1) 7 (2) 2

(3) 1 (4) 5

Answer (4)

Sol. $\lim\limits_{x\to 0} \dfrac{3+\alpha\sin x+\beta\cos x+\log_e(1-x)}{3\tan^2 x} = \dfrac{1}{3}$

As the limit is finite when $x \to 0$, denominator $3\tan^2 x \to 0$ so numerator has to be zero when $x \to 0$ to become $\frac{0}{0}$ form.

So, when $x \to 0$

$3 + \alpha\sin 0 + \beta\cos 0 + \log_e(1-0) = 0$

$3 + \beta = 0$

$\beta = -3$

As this is $\frac{0}{0}$ form, we apply L. Hopital rule here,

So, $\lim\limits_{x\to 0} \dfrac{\alpha\cos x - \beta\sin x + \frac{1(-1)}{1-x}}{6\tan x\sec^2 x} = \dfrac{1}{3}$



Again when $x \to 0$ denominator $6\tan x\sec^2 x \to 0$

So, numerator has to be zero

So, when $x \to 0$

$$\alpha\cos 0 - \beta\sin 0 - \frac{1}{1-0} = 0$$

$\Rightarrow \alpha - 1 = 0$

$\Rightarrow \alpha = 1$

Now,

$2\alpha - \beta = 2(1) - (-3)$

$= 2 + 3$

$= 5$

14. Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then the distance of the point $P(m, n)$ from the point $Q(-2, -3)$ is:

(1) 6 (2) 4

(3) 10 (4) 8

Answer (3)

Sol. As given, $2^m - 2^n = 56$

$\Rightarrow m = 6, n = 3$

$P(6, 3)$ and $Q(-2, -3)$

Distance $PQ = \sqrt{(6+2)^2 + (3+3)^2}$

$= 10$

15. Considering only the principal values of inverse trigonometric functions, the number of positive real values of x satisfying $\tan^{-1}(x) + \tan^{-1}(2x) = \frac{\pi}{4}$ is:

(1) More than 2 (2) 1

(3) 0 (4) 2

Answer (2)

Sol. $f(x) = \tan^{-1}x + \tan^{-1}2x$

$$f'(x) = \frac{1}{1+x^2} + \frac{2}{1+4x^2} > 0 \ \forall \ x \in R$$

$$\lim_{x\to\infty} f(x) = \pi$$

$$\lim_{x\to-\infty} f(x) = -\pi$$

Hence, $f(x)$ in monotonically increasing function for all $x \in R$ and range is $(-\pi, \pi)$.

Hence, $f(x) = \frac{\pi}{4}$ has exactly one solution for $x > 0$.

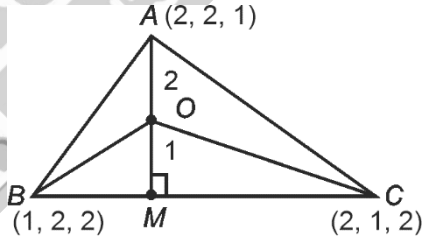
16. Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i} + 2\hat{j} + \hat{k}, \hat{i} + 2\hat{j} + 2\hat{k}$ and $2\hat{i} + \hat{j} + 2\hat{k}$ respectively. Let l_1, l_2 and l_3 be the lengths of perpendiculars drawn from the ortho center of the triangle on the sides AB, BC and CA respectively, then $l_1^2 + l_2^2 + l_3^2$ equals:

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$

(3) $\frac{1}{4}$ (4) $\frac{1}{5}$

Answer (1)

Sol.



$$|AB| = \sqrt{1^2 + 0^2 + (-1)^2} = \sqrt{2} = BC = CA$$

$\Rightarrow$ ABC is an equilateral triangle.

$\Rightarrow$ Orthocentre $\equiv$ centroid $\equiv$ circumcentre

Centroid $= \left(\frac{5}{3}, \frac{5}{3}, \frac{5}{3}\right) \equiv O$



And $M=\left(\frac{3}{2},\frac{3}{2},2\right)$ (Mid-point of BC)

$$|OM|=\sqrt{\left(\frac{5}{3}-\frac{3}{2}\right)^2+\left(\frac{5}{3}-\frac{3}{2}\right)^2+\left(\frac{5}{3}-2\right)^2}$$

$$=\sqrt{\frac{1}{36}+\frac{1}{36}+\frac{1}{9}}=\sqrt{\frac{6}{36}}=\frac{1}{\sqrt{6}}=I$$

$I_1 = I_2 = I_3 = I$

$$\Rightarrow \quad I_1^2+I_2^2+I_3^2=3I^2=3\times\frac{1}{6}=\frac{1}{2}$$

17. If $y = y(x)$ is the solution curve of the differential equation $(x^2 - 4)dy - (y^2 - 3y)dx = 0$, $x > 2$, $y(4) = \frac{3}{2}$ and the slope of the curve is never zero, then the value of $y(10)$ equals:

(1) $\frac{3}{1+(8)^{\frac{1}{4}}}$ (2) $\frac{3}{1+2\sqrt{2}}$

(3) $\frac{3}{1-(8)^{\frac{1}{4}}}$ (4) $\frac{3}{1-2\sqrt{2}}$

Answer (1)

Sol. $\left(x^2-4\right)dy-\left(y^2-3y\right)dx=0,\ x>2,\ y(4)=\frac{3}{2}$

$$\frac{dy}{y^2-3y}=\frac{dx}{x^2-4}$$

$$\Rightarrow \quad \frac{\ln|y-3|}{3}-\frac{\ln|y|}{3}=\frac{\ln|x-2|}{4}-\frac{\ln|x+2|}{4}+C$$

$$\Rightarrow \quad \frac{1}{3}\ln\left|\frac{y-3}{y}\right|=\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right|+\ln C,\ C>0$$

$$\Rightarrow \quad \left|\left(\frac{y-3}{y}\right)^{\frac{1}{3}}\right|\left|\left(\frac{x+2}{x-2}\right)^{\frac{1}{4}}\right|=K$$

$$y(4)=\frac{3}{2} \quad \Rightarrow \quad \left|\left(\frac{\frac{3}{2}-3}{\frac{3}{2}}\right)^{\frac{1}{3}}\right|\left|\frac{6}{2}\right|^{\frac{1}{4}}=K \quad \Rightarrow \quad K=3^{\frac{1}{4}}$$

$$\Rightarrow \quad \left|\left(\frac{y-3}{y}\right)^{\frac{1}{3}}\right|\left|\left(\frac{x+2}{x-2}\right)^{\frac{1}{4}}\right|=3^{\frac{1}{4}}$$

At $x = 10$, $\left|\left(\frac{y-3}{y}\right)^{\frac{1}{3}}\right|\left(\frac{12}{8}\right)^{\frac{1}{4}}=3^{\frac{1}{4}}$

$$\Rightarrow \quad \left(\frac{y-3}{y}\right)^{\frac{1}{3}}=2^{\frac{1}{4}}$$

$$\Rightarrow \quad \left|\frac{y-3}{y}\right|=2^{\frac{3}{4}}=8^{\frac{1}{4}}$$

$$\Rightarrow \quad \frac{3-y}{y}=8^{\frac{1}{4}}$$

$$\Rightarrow \quad y(10)=\frac{3}{1+8^{\frac{1}{4}}}$$

18. If $2\tan^2\theta - 5\sec\theta = 1$ has exactly 7 solutions in the interval $\left[0,\frac{n\pi}{2}\right]$, for the least value of $n \in N$, then $\sum_{k=1}^{n}\frac{k}{2^k}$ is equal to:

(1) $1-\frac{15}{2^{13}}$ (2) $\frac{1}{2^{13}}\left(2^{14}-15\right)$

(3) $\frac{1}{2^{14}}\left(2^{15}-15\right)$ (4) $\frac{1}{2^{15}}\left(2^{14}-14\right)$

Answer (2)



Sol. $2\tan^2\theta - 5\sec\theta - 1 = 0$

$\Rightarrow 2(\sec^2\theta - 1) - 5\sec\theta - 1 = 0$, $\sec\theta \in (-\infty, -1] \cup [1, \infty)$

$\Rightarrow 2\sec^2\theta - 5\sec\theta - 3 = 0$

$2\sec^2\theta - 6\sec\theta + \sec\theta - 3 = 0$

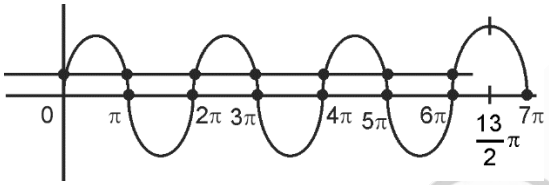
$2\sec\theta(\sec\theta - 3) + 1(\sec\theta - 3) = 0$

$\Rightarrow \sec\theta = \frac{-1}{2}$ or $\sec\theta = 3$

But $\sec\theta \in (-\infty, -1] \cup [1, \infty)$

$\Rightarrow \sec\theta = 3$

$\Rightarrow \cos\theta = \frac{1}{3}$



$\Rightarrow$ Least value of $n = 13$

$\Rightarrow \sum_{k=1}^{13} \frac{k}{2^k}$

$\Rightarrow$ A. G. P. $\Rightarrow S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \ldots + \frac{13}{2^{13}}$

$\Rightarrow \frac{S}{2} = \frac{1}{2^2} + \frac{2}{2^3} + \ldots + \frac{12}{2^{13}} + \frac{13}{2^{14}}$

$\Rightarrow S - \frac{S}{2} = \frac{1}{2} + \frac{1}{2^2} + \ldots + \frac{1}{2^{13}} - \frac{13}{2^{14}}$

$\Rightarrow \frac{S}{2} = \frac{\frac{1}{2}\left(1 - \left(\frac{1}{2}\right)^{13}\right)}{\left(1 - \frac{1}{2}\right)} = 1 - \frac{1}{2^{13}} - \frac{13}{2^{14}}$

$\Rightarrow S = 2 - \frac{1}{2^{12}} - \frac{13}{2^{13}} = \frac{2^{14} - 2 - 13}{2^{13}} = \frac{2^{14} - 15}{2^{13}}$

19. Let $\alpha = \frac{(4!)!}{(4!)^{3!}}$ and $\beta = \frac{(5!)!}{(5!)^{4!}}$. Then:

(1) $\alpha \in N$ and $\beta \in N$ (2) $\alpha \notin N$ and $\beta \in N$

(3) $\alpha \in N$ and $\beta \notin N$ (4) $\alpha \notin N$ and $\beta \notin N$

Answer (1)

Sol. $\alpha = \frac{(4!)!}{(4!)^{3!}}$ and $\beta = \frac{(5!)!}{(5!)^{4!}}$.

$4! = 24$

$\Rightarrow$ Divide 24 different objects into 6 persons of each

$\Rightarrow$ Numbers of ways of grouping

$= \frac{24!}{4!\cdot 4!\cdot 4!\ldots 4!} = \frac{24!}{(4!)^6} = \frac{(4!)!}{(4!)^6}$

$\Rightarrow \alpha \in N$

$5! = 120$

$\Rightarrow$ Divide 120 different into 24 persons 5 each

$\Rightarrow$ Number of ways of grouping

$= \frac{(5!)!}{(5!)^{24}} \Rightarrow \beta \in N$

$\Rightarrow \alpha, \beta$ both are natural number as they represent number of ways to group objects.

20. An urn contains 6 white and 9 black balls. Two successive draws of 4 balls are made without replacement. The probability, that the first draw gives all white balls and the second draw gives all black balls, is:

(1) $\frac{5}{256}$ (2) $\frac{5}{715}$

(3) $\frac{3}{715}$ (4) $\frac{3}{256}$

Answer (3)



Sol. 6 white, 9 black balls

First draw of 4 balls

$\Rightarrow$ Number of ways to get only white = $\left({}^{6}C_4\right)$

Second draw of 4 balls

$\Rightarrow$ Number of ways to get only black = ${}^{9}C_4$

Total cases $\Rightarrow$ First 4 balls out of 15 and then 4 balls out of remaining 11 balls

$\Rightarrow$ ${}^{15}C_4 \times {}^{11}C_4$

$\Rightarrow$ Probability = $\dfrac{{}^{6}C_4 \times {}^{9}C_4}{{}^{11}C_4 \cdot {}^{15}C_4} = \dfrac{3}{715}$

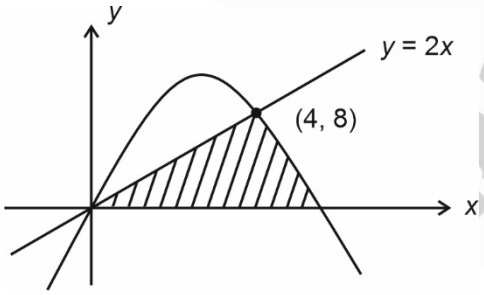
SECTION - B

Numerical Value Type Questions: This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

21. If the area of the region $\{(x, y) : 0 \le y \le \min\{2x, 6x - x^2\}\}$ is A, then 12 A is equal to _____.

Answer (304)

Sol. $\{(x, y) : 0 \le y \le \min\{2x, 6x - x^2\}\}$



Shaded region is required are.

Solving $y = 2x$ and $y = 6x - x^2$.

We will get (0, 0) or (4, 8)

$\therefore \quad A = \dfrac{1}{2} \times 4 \times 8 + \int_4^6 (6x - x^2)dx$

$= 16 + \left(3x^2 - \dfrac{x^3}{3}\right)_4^6$

$= 16 + \dfrac{28}{3}$

$\therefore \quad 12A = 304$

22. Let $f(x) = \int_0^x g(t)\log_e\left(\dfrac{1-t}{1+t}\right)dt$, where g is a continuous odd function.

If $\int_{-\pi/2}^{\pi/2}\left(f(x) + \dfrac{x^2\cos x}{1+e^x}\right)dx = \left(\dfrac{\pi}{\alpha}\right)^2 - \alpha$, then α is equal to _____.

Answer (2)

Sol. $f(x) = \int_0^x g(t)\log_e\left(\dfrac{1-t}{1+t}\right)dt$ is odd.

Now, $\int_{-\pi/2}^{\pi/2}\left(f(x) + \dfrac{x^2\cos x}{1+e^x}\right)dx$

$= \int_0^{\pi/2} \dfrac{(x^2\cos x)(1+e^x)}{(1+e^x)}dx$

$= \int_0^{\pi/2} x^2\cos x\,dx = \left[x^2\sin x\right]_0^{\pi/2} - 2\int_0^{\pi/2} x\sin x\,dx$

$= \dfrac{\pi^2}{4} - 2\times 1 = \dfrac{\pi^2}{4} - 2$

$\Rightarrow \alpha = 2$



23. Let the complex numbers α and $\frac{1}{\bar{\alpha}}$ lie on the circles $|z - z_0|^2 = 4$ and $|z - z_0|^2 = 16$ respectively, where $z_0 = 1 + i$. Then, the value of $100\,|\alpha|^2$ is _____.

Answer (20)

Sol. $|z - z_0|^2 = 4$ and $|z_0|^2 = 2$

$\Rightarrow (z - z_0)(\bar{z} - \bar{z}_0) = 4$

$\Rightarrow z\bar{z} - z_0\bar{z} - z_0\bar{z} + |z_0|^2 = 4$

$\Rightarrow |z_0|^2 - z_0\bar{z} - z_0\bar{z} - 2 = 0$

α lies on it

$\Rightarrow |\alpha|^2 - \alpha\bar{z}_0 - z_0\bar{\alpha} - 2 = 0$ ...(1)

Similarly,

$|z|^2 - z\bar{z}_0 - z_0\bar{z} - 14 = 0$

$\left[\because \left(\frac{1}{\bar{\alpha}}\right) = \frac{1}{|\bar{\alpha}|} = \frac{1}{|\alpha|^2}\right]$

$\frac{1}{\bar{\alpha}}$ lies on it

$\Rightarrow \frac{1}{|\alpha|^2} - \frac{1}{\bar{\alpha}}\cdot\bar{z}_0 - \bar{z}_0\frac{1}{\alpha} - 14 = 0$

$\Rightarrow 1 - \alpha\bar{z}_0 - z_0\bar{\alpha} - 14|\alpha|^2 = 0$ ...(2)

Using (1) and (2)

$\Rightarrow |\alpha|^2 - 2 = 1 - 14|\alpha|^2$

$\Rightarrow 15|\alpha|^2 = 3 \quad \Rightarrow |\alpha|^2 = \frac{1}{5}$

$\Rightarrow 100|\alpha|^2 = 20$

*** Note :** There is no such complex number α satisfy both circles.

24. If the sum of squares of all real values of α, for which the lines $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ and $\alpha x + 2y - 2 = 0$ **do not** form a triangle is p. then the greatest integer less than or equal to p is _____.

Answer (32)

Sol. $2x - y + 3 = 0$ ...(1)

$6x + 3y + 1 = 0$ ...(2)

$\alpha x + 2y - 2 = 0$ ...(3)

Triangle will not form if either at least two lines are parallel or lines are concurrent.

Case I

If two lines are parallel

$\frac{2}{\alpha} = \frac{-1}{2}$

$\Rightarrow \alpha = -4$

or

$\frac{6}{\alpha} = \frac{3}{2}$

$\Rightarrow \alpha = 4$

Case II

If lines are concurrent

$$\begin{vmatrix} 2 & -1 & 3 \\ 6 & 3 & 1 \\ \alpha & 2 & -2 \end{vmatrix} = 0$$

$\Rightarrow 8 - 10\alpha = 0$

$\Rightarrow \alpha = \frac{4}{5}$

$\Rightarrow \alpha_1^2 + \alpha_2^2 + \alpha_3^2 = p = 16 + 16 + \frac{16}{25}$

Greatest integer less than or equal to p is 32

25. The coefficient of x^{2012} in the expansion of $(1 - x)^{2008}(1 + x + x^2)^{2007}$ is equal to _____.

Answer (0)

Sol. Coefficient of x^{2012} in $(1 - x)^{2008}(1 + x + x^2)^{2007}$

$(1 - x)^{2008}(1 + x + x^2)^{2007}$

$= (1 - x)(1 - x^3)^{2007}$

$= (1 - x)\sum_{r=0}^{2007} {}^{2007}C_r(-x^3)^r$

$3r = 2012$ or $3r + 1 = 2012$

$r \notin \omega$ $r \notin \omega$

$\therefore$ No term contains x^{2012}

$\Rightarrow$ Coefficient of $x^{2012} = 0$



26. The lines $\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}$ and $\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}$ intersect at the point P. If the distance of P from the line $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$ is l, then $14l^2$ is equal to _____.

Answer (108)

Sol.

$$\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}=\lambda$$

$$\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}=t$$

$P(2\lambda+2,-2\lambda,16\lambda+7)$

$(4t-3,3t-2,t-2)$

$\Rightarrow 2\lambda+2=4t-3,\ -2\lambda=3t-2,\ 16\lambda+7=t-2$

$\Rightarrow \lambda=-\frac{1}{2},\ t=1$

$\Rightarrow P(1,1,-1)$

$\Rightarrow$

$P(1,1,-1)$

$A(-1,1,1)$ M $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$

$\overrightarrow{AP}=2\hat{i}-2\hat{k}$ $\qquad \vec{b}=2\hat{i}+3\hat{j}+\hat{k}$,

$AP^2=4+4=8$

$AM=\left|\overrightarrow{AP}\cdot\hat{b}\right|=\frac{2}{\sqrt{14}} \Rightarrow AM^2=\frac{4}{14}=\frac{2}{7}$

$\Rightarrow PM=\sqrt{8-\frac{2}{7}}=\sqrt{\frac{54}{7}}=\ell$

$\Rightarrow \frac{54}{7}=\ell^2 \Rightarrow 14\ell^2=108$

27. The mean and standard deviation of 15 observations were found to be 12 and 3 respectively. On rechecking it was found that an observation was read as 10 in place of 12. If μ and σ^2 denote the mean and variance of the correct observations respectively, then $15(\mu+\mu^2+\sigma^2)$ is equal to _____.

Answer (2521)

Sol. $\bar{x}=12,\ \sigma_1=3$

$$\Rightarrow \frac{x_1+x_2+\cdots+x_{14}+10}{15}=12$$

$$\Rightarrow x_1+x_2+\cdots+x_{14}=170$$

$$\frac{(x_1-12)^2+(x_2-12)^2+\cdots+(x_{14}-12)^2+(10-12)^2}{15}=9$$

$$\mu=\frac{x_1+\cdots+x_{14}+12}{15}=\frac{170+12}{15}=\frac{182}{15}=12+\frac{2}{15}$$

$$\Rightarrow \sigma^2=\frac{\left(x_1-12-\frac{2}{15}\right)^2+\cdots+\left(x_{14}-12-\frac{2}{15}\right)^2+\left(12-12-\frac{2}{15}\right)^2}{15}$$

$$15\sigma^2=(x_1-12)^2+\cdots+(x_{14}-12)^2+14\times\frac{4}{225}-\frac{4}{15}$$

$$\sum_{r=1}^{14}(x_r-12)$$

$$=131+\frac{56}{225}-\frac{4}{15}(170-168)$$

$$=131+\frac{56}{225}-\frac{8}{15}=131-\frac{64}{225}$$

$\Rightarrow 15(\mu+\mu^2+\sigma^2)=2521$

28. Let A be a 2×2 real matrix and I be the identify matrix of order 2. If the roots of the equation $|A-xI|=0$ be -1 and 3, then the sum of the diagonal elements of the matrix A^2 is _____.

Answer (10)



Sol. Let $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$|A-xI|=0 \Rightarrow \begin{vmatrix} a-x & b \\ c & d-x \end{vmatrix}=0$$

$(a-x)(d-x)-bc=0$

$x^2-(a+d)x+ad-bc=0$

$\Rightarrow a+d=2,\ ad-bc=-3$

$$A^2=\begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix}$$

$\Rightarrow$ Sum $= a^2+d^2+2bc=(a+d)^2-2ad+2bc$

$= 4-2(-3)=10$

29. Consider a circle $(x-\alpha)^2+(y-\beta)^2=50$, where α, $\beta>0$. If the circle touches the line $y+x=0$ at the point P, whose distance from the origin is $4\sqrt{2}$, then $(\alpha+\beta)^2$ is equal to _____.

Answer (100)

Sol. Let $P\left(\alpha+5\sqrt{2}\cos\theta,\ \beta+5\sqrt{2}\sin\theta\right)$

Equation of tangent to circle $(x-\alpha)^2+(y-\beta)^2=50$ at point P is

$(x-\alpha)\cos\theta+(y-\beta)\sin\theta=5\sqrt{2}$

$\Rightarrow x\cos\theta+y\sin\theta=\alpha\cos\theta+\beta\sin\theta+5\sqrt{2}$

$x+y=0$

$$\Rightarrow \frac{\cos\theta}{1}=\frac{\sin\theta}{1}=\frac{\alpha\cos\theta+\beta\sin\theta+5\sqrt{2}}{0}$$

$\Rightarrow \tan\theta=1,\ \alpha\cos\theta+\beta\sin\theta+5\sqrt{2}=0$

$$\tan\theta=1 \Rightarrow \sin\theta=\frac{1}{\sqrt{2}},\ \cos\theta=\frac{1}{\sqrt{2}}$$

Or $\sin\theta=\cos\theta=-\frac{1}{\sqrt{2}}$

$\Rightarrow (\alpha+\beta)\pm 10$

$\Rightarrow (\alpha+\beta)^2=100$

30. If the solution curve, of the differential equation $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ passing through the point (2, 1) is

$$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{\beta}\log_e\left(\alpha+\left(\frac{y-1}{x-1}\right)^2\right)=\log_e|x-1|,$$

then $5\beta+\alpha$ is equal to.

Answer (11)

Sol. $\frac{dy}{dx}=\frac{x+y-2}{x-y}$

$x+y-2=0$ and $x-y=0$ intersect at (1, 1)

$x=X+1,\ y=Y+1$

$$\Rightarrow \frac{dY}{dX}=\frac{X+Y}{X-Y}$$

Put $Y=tX \rightarrow \frac{dY}{dX}=t+X\frac{dt}{dX}$

$$\Rightarrow t+X\frac{dt}{dX}=\frac{1+t}{1-t}$$

$$X\frac{dt}{dX}=\frac{1+t^2}{1-t}$$

$$\Rightarrow \frac{dX}{X}=\frac{1-t}{1+t^2}dt$$

$$\ln|X|=\tan^{-1}t-\frac{1}{2}\ln|1+t^2|$$

$$\ln|x-1|=\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ln\left|1+\left(\frac{y-1}{x-1}\right)^2\right|+C$$

Passes through (2, 1) $\Rightarrow C=0$

$$\Rightarrow \tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ln\left(1+\left(\frac{y-1}{x-1}\right)^2\right)=\ln|x-1|$$

$\Rightarrow 5\beta+\alpha=11$



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

31. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The property of body, by virtue of which it tends to regain its original shape when the external force is removed, is Elasticity.

Reason (R) : The restoring force depends upon the bonded inter atomic and inter molecular force of solid.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(2) **(A)** is false but **(R)** is true

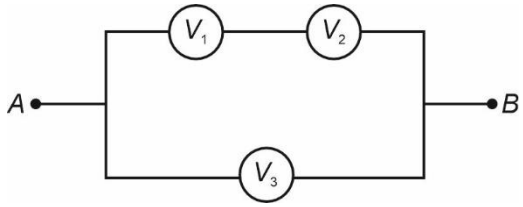
(3) **(A)** is true but **(R)** is false

(4) Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**

Answer (1)

Sol. The elasticity is responsible for shape regain of material and it cause due to inter atomic bond and intermolecular bond.

32. Three voltmeters, all having different internal resistances are joined as shown in figure. When some potential difference is applied across A and B, their readings are V_1, V_2 and V_3. Choose the correct option.



(1) $V_1 + V_2 > V_3$ (2) $V_1 = V_2$

(3) $V_1 \neq V_3 - V_2$ (4) $V_1 + V_2 = V_3$

Answer (4)

Sol. V_1 and V_2 are in series and they are in parallel with V_3.

$\therefore \quad V_1 + V_2 = V_3$

33. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Work done by electric field on moving a positive charge on an equipotential surface is always zero.

Reason (R) : Electric lines of forces are always perpendicular to equipotential surfaces.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **(A)** is correct but **(R)** is not correct

(2) **(A)** is not correct but **(R)** is correct

(3) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(4) Both **(A)** and **(R)** are correct but **(R)** is **not** the correct explanation of **(A)**

Answer (3)

Sol. The potential at equipotential surface remain same everywhere, so no work is needed to move any charged particle and $\vec{E}$ is perpendicular to equipotential surface.

